EXPERIMENT 9

CO4 LAB-9

Program:

```
from itertools import permutations
cost = [[0, 10, 15],
        [10, 0, 20],
        [15, 20, 0]]
n = len(cost)
best = 999
for p in permutations(range(1, n)):
    path = (0,) + p + (0,)
    total = sum(cost[path[i]][path[i+1]] for i in range(n))
    if total < best:
        best = total
        route = path
print("Minimum cost:", best)
print("Path:", route)
```

Output:

```
Minimum cost: 45
Path: (0, 1, 2, 0)
```